R21

Code No.: U21PC602CS

M V S R ENGINEERING COLLEGE (Autonomous)

B.E. VI Semester (Supple.) (Branch: CSE, CSE (IOT-CS-BCT)) Examination, February 2026

ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

Time : 3 hours Max. Marks : 70

Note : i) FIRST Question is compulsory and answer any **FOUR** questions from the remaining six questions. Each question carries 14 Marks.
ii) Answers to each question must be written at one place only and in the same order as they occur in the question paper.
iii) Missing data, if any, may suitably be assumed.

Q.No.			Marks	CO	BT levels
1.	a)	What is heuristic function?	(2)	1	1
	b)	List the elements of propositional logic.	(2)	2	1
	c)	What is Semi Supervised learning?	(2)	3	1
	d)	Define Multilayer neural network?	(2)	4	1
	e)	What is a Gaussian mixture model?	(2)	5	1
	f)	List the Applications of Expert System.	(2)	2	1
	g)	What is Recall?	(2)	4	1
2	(a)	Discuss about A* algorithm with Suitable Example.	(7)	1	2
	(b)	Elaborate alpha-beta pruning with Suitable Example.	(7)	1	2
3	(a)	Explain in detail the connectives used in propositional logic.	(7)	2	2
	(b)	Explain different phases in Building Expert System.	(7)	2	2
4	(a)	Discuss the Steps to find the line of Best Fit by using the Least Squares Method	(7)	3	6
	(b)	Demonstrate KNN Algorithm with example.	(7)	3	2
5	(a)	Explain in detail back propagation Algorithm.	(7)	4	4
	(b)	Comparison between K-Fold Cross-Validation and Hold Out Method	(7)	4	2
6	(a)	Distinguish between types of hierarchical clustering techniques.	(7)	5	4
	(b)	Discuss Reinforcement Learning Example	(7)	5	2
7	(a)	Calculate Conditional Probability of Events in a Bayesian Network	(5)	2	5
	(b)	Why is Non-Linearity Important in Neural Networks?	(5)	4	1
	(c)	How to Implement Speech Recognition	(4)	5	1

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